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Education

Ph.D., Purdue University, 2009.

M.S. Mechanical Engineering, Texas Tech University, 2004.

B.S. Mechanical Engineering, *Cum Laude*, Texas Tech University, 2002.

Professional Registration

Professional Engineer, Texas, #118233

Professional Appointments

Professor, Hildebrand Department of Petroleum and Geosystems Engineering,
The University of Texas at Austin,
August 2024–Present.

Professor (by courtesy), Department of Aerospace Engineering
and Engineering Mechanics,
The University of Texas at Austin,
August 2024–Present.

George H. Fancher Professorship in Petroleum Engineering,
The University of Texas at Austin,
September 2016–Present.

Core Faculty, Oden Institute for Computational Engineering and Sciences,
The University of Texas at Austin,
November 2017–Present.

Associate Professor, Hildebrand Department of Petroleum and Geosystems Engineering,
The University of Texas at Austin,
September 2017–August 2024.

Associate Professor (by courtesy), Department of Aerospace Engineering
and Engineering Mechanics,
The University of Texas at Austin,
September 2017–Present.

Affiliated Faculty, Institute for Computational Engineering and Sciences,
The University of Texas at Austin,
December 2014–October 2017.

Assistant Professor, Department of Petroleum and Geosystems Engineering,
The University of Texas at Austin,
August 2014–August 2017.

Assistant Professor (by courtesy), Department of Aerospace Engineering
and Engineering Mechanics,
The University of Texas at Austin,
January 2014–August 2017.

Assistant Professor, Department of Mechanical Engineering,
The University of Texas at San Antonio,
August 2011–August 2014.

Adjunct Associate Professor, Department of Mechanical Engineering
The University of New Mexico,
September 2010–May 2011.

Senior Member of the Technical Staff, Terminal Ballistics Technology Department,
Sandia National Laboratories
January 2006–August 2011.

Member of the Technical Staff, Terminal Ballistics Technology Department,
Sandia National Laboratories
August 2004–January 2006.

Awards & Honors

2023 SPE Regional Data Science and Engineering Analytics Award

2018 ICES W.A. "Tex" Moncrief Grand Challenge Award

2015 SPE Petroleum Engineering Innovative Teaching Award

2013 Air Force Office of Scientific Research Young Investigator Award

2013 '40 Under 40' - San Antonio Business Journal

Consulting

Hilcorp AK, October 2023 – Present

Injection scheduling and optimization

Bazean Corp., January 2019 – August 2019

Executive in residence

Curl, Stahl, Geis, November 2014 – February 2015

Engineering analysis and expert witness testimony at trial

Refereed Journal Articles

Published

65. **J.T. Foster**, S.T. Tirmizi, E. Ikpesu, and M. Hesse. A finite deformation formulation for reacting porous media with volume change. *Journal of the Mechanics and Physics of Solids*, 217:106811, 2026. doi:10.1016/j.jmps.2026.106811.

64. E. Rustamzade, **J.T. Foster**, and M.J. Pyrcz. Machine Learning Proxy Modeling for Production Forecasting using Ridge-Regularized Polynomial Regression for Deepwater Commingled Reservoirs. *SPE Journal*, 31(8):5116–5126, 2026. doi:10.2118/233792-pa.

63. D. Wang, **J.T. Foster**, Y. Lu, E. Rustamzade, H. Xiong, A. Thompson, and M.J. Pyrcz. A comprehensive machine learning workflow for classifying production profiles in unconventional reservoirs. *Energy Exploration & Exploitation*, 44(4):1806–1829, 2026. doi:10.1177/01445987261433779.

62. **J.T. Foster**. The Rancher, the Hunter, the Renaissance Man: A Tribute to my Beloved Friend Tinsley Oden. *Computing in Science & Engineering*, 28(1):10–14, 2026. doi:10.1109/mcse.2026.3667292.

61. Ahmed Merzoug, Fehmi Özbayrak, John T Foster, and Michael J Pyrcz. Beyond random forest: how spatial bagging and spatial random forest dominate for subsurface applications? *Computational Geosciences*, 29(6):52, 2025.

60. F. Özbayrak, **J.T. Foster**, and M.J. Pyrcz. Spatial bagging for predictive machine learning uncertainty quantification. *Computers & Geosciences*, 203:105947, 2025. doi:10.1016/j.cageo.2025.105947.

59. **J.T. Foster** and X. Xu. Revisiting finite deformation poromechanics: Deriving a nonlinear Biot coefficient from first principles. *Journal of the Mechanics and Physics of Solids*, 204:105259, 2025. doi:10.1016/j.jmps.2025.105259.

58. M.M. Akmal, K. Sepehrnoori, **J.T. Foster**, and M.J. Pyrcz. Modelling of joule-thomson cooling effect using a modified shift-deeponet method for predicting hydrate onset during co2 sequestration. *Geoenergy Science and Engineering*, 243:213320, 2024. doi:10.1016/j.geoen.2024.213320.

57. F. Özbayrak, **J.T. Foster**, and M.J. Pyrcz. Spatial bagging to integrate spatial correlation into ensemble machine learning. *Computers & Geosciences*, page 105558, 2024. doi:10.1016/j.cageo.2024.105558.

56. H. You, X. Xu, Yue Yu, S.A. Silling, M D'Elia, and **J.T. Foster**. Towards a unified nonlocal, peridynamics framework for the course-graining of molecular dynamics data with fractures. *Applied Mathematics and Mechanics*, (44):1125–1150, 2023. doi:/10.1007/s10483-023-2996-8.

55. E. Rustamzade, W. Pan, **J.T. Foster**, and M.J. Pyrcz. Comparison of commingled and sequential production schemes by sensitivity analysis for gulf of mexico paleogene deepwater. *Energy Exploration & Exploitation*, 2023. `doi:10.1177/01445987231195679`.

54. D.J. Littlewood, M.L. Parks, **J.T. Foster**, J.A. Mitchell, and P. Diehl. The Peridigm Meshfree Peridynamcis Code. *Journal of Peridynamics and Nonlocal Modeling*, 2023. `doi:10.1007/s42102-023-00100-0`.

53. E. Maldonado-Cruz, **J.T. Foster**, and M.J. Pyrcz. Sonic Well-Log Imputation Through Machine-Learning-Based Uncertainty Models. *Petrophysics*, 64(02):253–270, 04 2023. `doi:10.30632/PJV64N2-2023a7`.

52. M. Yang and **J.T. Foster**. Using physics-informed neural networks to solve for permeability field under two-phase flow in heterogeneous porous media. *Journal of Machine Learning for Modeling and Computing*, 2023. `doi:10.1615/JMachLearnModelComput.2023046921`.

51. M.B. Abdullah, M. Delshad, K. Sepehrnoori, M.T. Balhoff, **J.T. Foster**, and M.T. Al-Murayri. Physics-Based and Data-Driven Polymer Rheology Model. *SPE Journal*, pages 1–23, 02 2023. `doi:10.2118/214307-PA`.

50. X. Xu and **J.T. Foster**. The peridynamic Jacobian. *Journal of Peridynamics and Nonlocal Modeling*, 2022. `doi:10.1007/s42102-022-00091-4`.

49. X. Xu, M. D'Elia, C. Glusa, and **J.T. Foster**. Machine-learning of nonlocal kernels for anomalous subsurface transport from breakthrough curves. *Numerical Algebra, Control and Optimization*, 0, 2022. `doi:10.3934/naco.2022025`.

48. Y. Bazilevs, M. Behzadinasab, and **J.T. Foster**. Simulating concrete failure using the Microplane (M7) constitutive model in correspondence-based peridynamics: Validation for classical fracture tests and extension to discrete fracture. *Journal of the Mechanics and Physics of Solids*, page 104947, 2022. `doi:10.1016/j.jmps.2022.104947`.

47. M. Yang and **J.T. Foster**. Multi-output physics-informed neural networks for forward and inverse pde problems with uncertainties. *Computer Methods in Applied Mechanics and Engineering*, page 115041, 2022. `doi:10.1016/j.cma.2022.115041`.

46. Y. Leng, X. Tian, L. Demkowicz, H. Gomez, and **J.T. Foster**. A Petrov-Galerkin method for nonlocal convection-dominated diffusion problems. *Journal of Computational Physics*, 452:110919, 2022. `doi:10.1016/j.jcp.2021.110919`.

45. M. Behzadinasab, G. Moutsanidis, N. Trask, **J.T. Foster**, and Y. Bazilevs. Coupling of iga and peridynamics for air-blast fluid-structure interaction using an immersed approach. *Forces in Mechanics*, 4:100045, 2021. `doi:10.1016/j.finmec.2021.100045`.

44. X. Xu, C. Glusa, M. D'Elia, and **J.T. Foster**. A FETI approach to domain decomposition for meshfree discretizations of nonlocal problems. *Computer Methods in Applied Mechanics and Engineering*, 387:114148, 2021. `doi:10.1016/j.cma.2021.114148`.

43. X. Xu, M. D'Elia, and **J.T. Foster**. A machine-learning framework for peridynamic material models with physical constraints. *Computer Methods in Applied Mechanics and Engineering*, 386, December 2021. doi:10.1016/j.cma.2021.114062.

42. M. Yang and **J.T. Foster**. hp-variational physics-informed neural networks for nonlinear two-phase transport in porous media. *Journal of Machine Learning for Modeling and Computing*, 2:15–32, 2021. doi:10.1615/JMachLearnModelComput.2021038005.

41. S. Agrawal, J.R. York, **J.T. Foster**, and M.M. Sharma. Coupling Peridynamics with the Classical Methods for Modeling Hydraulic Fracture Growth in Heterogeneous Reservoirs. *SPE Journal*, pages 1–19, 04 2021. doi:10.2118/205393-PA.

40. Y. Leng, X. Tian, N.A. Trask, and **J.T. Foster**. Asymptotically Compatible Reproducing Kernel Collocation and Meshfree Integration for Nonlocal Diffusion. *SIAM Journal on Numerical Analysis*, 59(1):88–118, 2021. doi:10.1137/19M1277801.

39. M. Behzadinasab, **J.T. Foster**, and Y. Bazilevs. A Unified, Stable, and Accurate Meshfree Framework for Peridynamic Correspondence Modeling—Part II: Wave Propagation and Enforcement of Stress Boundary Conditions. *Journal of Peridynamics and Nonlocal Modeling*, 2020. doi:10.1007/s42102-020-00039-6.

38. Y. Leng, X. Tian, N.A. Trask, and **J.T. Foster**. Asymptotically compatible reproducing kernel collocation and meshfree integration for the peridynamic navier equation. *Computer Methods in Applied Mechanics and Engineering*, 370:113264, 2020. doi:10.1016/j.cma.2020.113264.

37. X. Xu and **J.T. Foster**. Deriving peridynamic influence functions for one-dimensional elastic materials with periodic microstructure. *Journal of Peridynamics and Nonlocal Modeling*, 2020. doi:10.1007/s42102-020-00037-8.

36. M. Behzadinasab and **J.T. Foster**. Revisiting the third sandia fracture challenge: a bond-associated, semi-lagrangian peridynamic approach to modeling large deformation and ductile fracture. *International Journal of Fracture*, 2020. doi:10.1007/s10704-020-00455-1.

35. S. Agrawal, S. Zheng, **J.T. Foster**, and M.M. Sharma. Coupling of meshfree peridynamics with the finite volume method for poroelastic problems. *Journal of Petroleum Science and Engineering*, page 107252, 2020. doi:10.1016/j.petrol.2020.107252.

34. M. Behzadinasab and **J.T. Foster**. A semi-lagrangian constitutive correspondence framework for peridynamics. *Journal of the Mechanics and Physics of Solids*, 137:103862, April 2020. doi:10.1016/j.jmps.2019.103862.

33. A. Katiyar, S. Agrawal, H. Ouchi, P. Seleson, **J.T. Foster**, and M.M. Sharma. A general peridynamics model for multiphase transport of non-Newtonian compressible fluids in porous media. *Journal of Computational Physics*, 2019. doi:10.1016/j.jcp.2019.109075.

32. M. Behzadinasab and **J.T. Foster**. On the stability of the generalized, finite deformation correspondence model of peridynamics. *International Journal of Solids and Structures*, 2019. doi:10.1016/j.ijsolstr.2019.07.030.

31. S.L.B. Kramer, A. Jones, A. Mostafa, B. Ravaji, T. Tancogne-Dejean, C.C. Roth, M. Gorji Bandpay, K. Pack, **J.T. Foster**, M. Behzadinasab, and others. The third Sandia Fracture Challenge: predictions of ductile fracture in additively manufactured metal. *International Journal of Fracture*, June 2019. doi:10.1007/s10704-019-00361-1.

30. M. Behzadinasab and **J.T. Foster**. The third Sandia Fracture Challenge: peridynamic blind prediction of ductile fracture characterization in additively manufactured metal. *International Journal of Fracture*, June 2019. doi:10.1007/s10704-019-00363-z.

29. Y. Leng, X. Tian, and **J.T. Foster**. Super-convergence of reproducing kernel approximation. *Computer Methods in Applied Mechanics and Engineering*, 352:488–507, August 2019. doi:10.1016/j.cma.2019.04.038.

28. D. Kamensky, M. Behzadinasab, **J.T. Foster**, and Y. Bazilevs. Peridynamic modeling of frictional contact. *Journal of Peridynamics and Nonlocal Modeling*, Apr 2019. doi:10.1007/s42102-019-00012-y.

27. M. Behzadinasab, T.J. Vogler, A.M. Peterson, R. Rahman, and **J.T. Foster**. Peridynamics modeling of a shock wave perturbation decay experiment in granular materials with intra-granular fracture. *Journal of Dynamic Behavior of Materials*, 4(4):529–542, December 2018. doi:10.1007/s40870-018-0174-2.

26. M. Pasetto, Y. Leng, J.S. Chen, **J.T. Foster**, and P. Seleson. A reproducing kernel enhanced approach for peridynamic solutions. *Computer Methods in Applied Mechanics and Engineering*, 340:1044–1078, October 2018. doi:10.1016/j.cma.2018.05.010.

25. **J.T. Foster** and X. Xu. A generalized, ordinary, finite deformation constitutive correspondence model for peridynamics. *International Journal of Solids and Structures*, 141–142:245–253, June 2018. doi:10.1016/j.ijsolstr.2018.02.026.

24. H. Ouchi, A. Katiyar, **J.T. Foster**, and M. M. Sharma. A peridynamics model for the propagation of hydraulic fractures in naturally fractured reservoirs. *SPE Journal*, Preprint(SPE-173361-PA), May 2017. doi:10.2118/173361-PA.

23. H. Ouchi, **J.T. Foster**, and M.M. Sharma. Effect of reservoir heterogeneity on the vertical migration of hydraulic fractures. *Journal of Petroleum Science and Engineering*, 151:384–408, 2017. doi:10.1016/j.petrol.2016.12.034.

22. J.T. O'Grady and **J.T. Foster**. A meshfree method for bending and failure in non-ordinary peridynamic shells. *Computational Mechanics*, 57(6):921–929, June 2016. doi:10.1007/s00466-016-1269-z.

21. **J.T. Foster**. A variationally consistent approach to constrained motion. *ASME. J. Appl. Mech.*, 83(5), May 2016. doi:10.1115/1.4032856.

20. R. Rahman and **J.T. Foster**. Onto resolving spurious wave reflection problem with changing nonlocality among various length scales. *Communications in Nonlinear Science and Numerical Simulation*, 34:86–122, 2016. doi:10.1016/j.cnsns.2015.10.003.

19. R. Rahman and **J.T. Foster**. Peridynamic theory of solids from the perspective of classical statistical mechanics. *Physica-A*, 437:162–183, November 2015. doi:10.1016/j.physa.2015.05.099.

18. R. Rahman and **J.T. Foster**. A molecular dynamics based investigation of thermally vibrating graphene under different boundary conditions. *Physica E: Low-dimensional Systems and Nanostructures*, 72:25–47, August 2015. doi:10.1016/j.physe.2015.04.007.

17. H. Ouchi, A. Katiyar, J.R. York, **J.T. Foster**, and M.M. Sharma. A fully coupled porous flow and geomechanics model for fluid driven cracks: a peridynamics approach. *Computational Mechanics*, 55(3):561–576, March 2015. doi:10.1007/s00466-015-1123-8.

16. J.T. O'Grady and **J.T. Foster**. Peridynamic plates and flat shells: A non-ordinary state-based model. *International Journal of Solids and Structures*, 51(25–26):4572–4579, 2014. doi:10.1016/j.ijsolstr.2014.09.003.

15. J.T. O'Grady and **J.T. Foster**. Peridynamic beams: A non-ordinary state-based model. *International Journal of Solids and Structures*, 51(18):3177–3183, 2014. doi:10.1016/j.ijsolstr.2014.05.014.

14. M. Bessa, **J.T. Foster**, T. Belytschko, and W.K. Liu. A meshfree unification: Reproducing kernel peridynamics. *Computational Mechanics*, 53(6):1251–1264, 2014. doi:10.1007/s00466-013-0969-x.

13. M.D. Brothers, **J.T. Foster**, and H.R. Millwater. A comparison of different methods for calculating tangent-stiffness matrices in a massively parallel computational peridynamics code. *Computer Methods in Applied Mechanics and Engineering*, 279:247—267, September 2014. doi:10.1016/j.cma.2014.06.034.

12. R. Rahman and **J.T. Foster**. Bridging the length scales through nonlocal hierarchical multiscale modeling scheme. *Computational Material Science*, 92:401–415, September 2014. doi:10.1016/j.commatsci.2014.05.052.

11. R. Rahman, **J.T. Foster**, and A. Haque. A multiscale modeling scheme based on peridynamic theory. *International Journal of Multiscale Computational Engineering*, 12(3):223–248, 2014. doi:10.1615/IntJMultCompEng.2014007954.

10. R. Rahman and **J.T. Foster**. Deformation mechanism of graphene in amorphous polyethylene: A molecular dynamics based study. *Computational Material Science*, 87:232–240, May 2014. doi:10.1016/j.commatsci.2014.02.023.

9. A. Katiyar, **J.T. Foster**, H. Ouchi, and M.M. Sharma. A peridynamic formulation of pressure driven convective fluid transport in porous media. *Journal of Computational Physics*, 261:209–229, March 2014. doi:10.1016/j.jcp.2013.12.039.

8. E.E. Nishida, **J.T. Foster**, and P.E. Briseno. Constant-strain-rate testing of a G10 laminate composite through optimized Kolsky bar pulse shaping techniques. *Journal of Composite Materials*, 47(23):2895–2903, 2013. doi:10.1177/0021998312460263.

7. R. Rahman, **J.T. Foster**, and A. Haque. Molecular dynamics simulation and characterization of graphene-cellulose nanocomposites. *The Journal of Physical Chemistry A*, 117(25):5344–5353, 2013. doi:10.1021/jp402814t.

6. **J.T. Foster**. Comments on the validity of test conditions in Kolsky bar experiments of elastic-brittle materials. *Experimental Mechanics*, 52(9):1559–1563, Brief Technical Note 2012. doi:10.1007/s11340-012-9592-6.

5. **J.T. Foster**, D.J. Frew, M.J. Forrestal, E.E. Nishida, and W. Chen. Shock testing accelerometers with a Hopkinson pressure bar. *International Journal of Impact Engineering*, 46:56–61, August 2012. doi:10.1016/j.ijimpeng.2012.02.006.

4. **J.T. Foster**, S.A. Silling, and W. Chen. An energy based failure criterion for use with peridynamic states. *International Journal of Multiscale Computational Engineering*, 9(6):675–688, 2011. doi:10.1615/IntJMultCompEng.2011002407.

3. **J.T. Foster**, W. Chen, and V.K. Luk. Dynamic crack initiation toughness of 4340 steel at constant loading rates. *Engineering Fracture Mechanics*, 78(6):1264 – 1276, 2011. doi:10.1016/j.engfracmech.2011.02.019.

2. **J.T. Foster**, S.A. Silling, and W.W. Chen. Viscoplasticity using peridynamics. *International Journal for Numerical Methods in Engineering*, 81(10):1242–1258, 2010. doi:10.1002/nme.2725.

1. **J.T. Foster**, A.A. Barhorst, C.N. Wong, and M.T. Bement. Modeling Loose Joints in Elastic Structures–Experimental Results and Validation. *Journal of Vibration and Control*, 15(4):549–565, 2009. doi:10.1177/1077546307082908.

Books Authored

1. **J.T. Foster**. *Introduction to High-Performance Computing*. e-book, 2022. URL: https://johnfoster.pge.utexas.edu/hpc-book

2. **J.T. Foster**. *Numerical Methods and Computer Programming*. e-book, 2019. URL: https://johnfoster.pge.utexas.edu/numerical-methods-book

Books Edited

1. F. Bobaru, **J.T. Foster**, P.H. Geubelle, and S.A. Silling, editors. *Handbook of Peridynamic Modeling*. Number ISBN 9781482230437 in Advances in Applied Mathematics. Chapman and Hall/CRC Press, 2016

Book Chapters

1. **J.T. Foster**. *Handbook of Peridynamic Modeling*, chapter Constitutive Modeling in Peridynamics. Number ISBN 9781482230437 in Advances in Applied Mathematics. Chapman and Hall/CRC Press, 2016

Conference Proceedings

17. S. Agrawal, J. York, **J.T. Foster**, and M. Sharma. Coupling meshfree peridynamics with the classical methods for modeling hydraulic fracture growth in heterogeneous reservoirs. In *Unconventional Resources Technology Conference, 20–22 July 2020*, pages 180–203. Unconventional Resources Technology Conference (URTEC), 2020.

16. M. Behzadinasab, T.J. Vogler, and **J.T. Foster**. Modeling perturbed shock wave decay in granular materials with intra-granular fracture. In *20th Biennial APS Conference on Shock Compression of Condensed Matter*, 2017.

15. H. Ouchi, S. Agrawal, **J.T. Foster**, and M.M. Sharma. Effect of small scale heterogeneity on the growth of hydraulic fractures. In *SPE Hydraulic Fracturing Technology Conference and Exhibition*. Society of Petroleum Engineers, 2017. `doi:10.2118/184873-MS`.

14. E.A. Lynd, **J.T. Foster**, and Q.P. Nguyen. An application of the Isogeometric Analysis Method to reservoir simulation. In *78th EAGE Conference and Exhibition*, number SPE-180110-MS. Society of Petroleum Engineers, 2016. `doi:10.2118/180110-MS`.

13. H. Ouchi, A. Katiyar, **J.T. Foster**, and M.M. Sharma. A Peridynamics Model for the Propagation of Hydraulic Fractures in Heterogeneous, Naturally Fractured Reservoirs. In *SPE Hydraulic Fracturing Technology Conference*, number SPE-173361-MS. Society of Petroleum Engineers, February 2015. `doi:10.2118/173361-MS`.

12. J.T. O'Grady and **J.T. Foster**. Peridynamic beams and plates: A non-ordinary state-based model. In *ASME 2014 International Mechanical Engineering Congress and Exposition*, number IMECE2014-39887, 2014. `doi:10.1115/IMECE2014-39887`.

11. J.R. York, **J.T. Foster**, E.E. Nishida, and B. Song. A novel torsional Kolksy bar for constant-strain-rate materials testing. In B. Song, D. Casem, and J. Kimberly, editors, *Dynamic Behavior of Materials, Volume 1*, Conference Proceedings of the Society for Experimental Mechanics Series. Springer New York, 2013. `doi:10.1007/978-3-319-00771-7_36`.

10. **J.T. Foster** and E.E. Nishida. *A priori* pulse shaper design for constant-strain-rate tests of elastic-brittle materials. In V. Chalivendra, B. Song, and D. Casem, editors, *Dynamic Behavior of Materials, Volume 1*, Conference Proceedings of the Society for Experimental Mechanics Series, pages 379–386. Springer New York, 2013. `doi:10.1007/978-1-4614-4238-7_49`.

9. **J.T. Foster**, D.J. Frew, M.J. Forrestal, E.E. Nishida, and W. Chen. Shock testing accelerometers with a Hopkinson pressure bar. *Experimental and Applied Mechanics, Volume 6*, pages 229–237, 2011. `doi:10.1007/978-1-4614-0222-0_29`.

8. **J.T. Foster**, W. Chen, and V.K. Luk. Dynamic fracture initiation toughness of high strength steel alloys. In *DYMAT 2009-9th International Conference on the Mechanical and Physical Behaviour of Materials under Dynamic Loading*, volume 1, pages 407–412, 2009. doi:10.1051/dymat/2009058.

7. **J.T. Foster**, S.A. Silling, and W.W. Chen. State based peridynamic modeling of dynamic fracture. In *DYMAT 2009-9th International Conference on the Mechanical and Physical Behaviour of Materials under Dynamic Loading*, volume 2, pages 1529–1535, 2009. doi:10.1051/dymat/2009216.

6. **J.T. Foster**, S.A. Silling, and W.W. Chen. State based peridynamic modeling of dynamic fracture. In *SEM 2009 Conference on Experimental and Applied Mechanics*, number 33. SEM, 2009.

5. **J.T. Foster**, V.K. Luk, and W. Chen. Dynamic initiation fracture toughness of high strength steel alloys. In *Proceedings of the XIth International Congress and Exposition. Orlando, Florida Society for Experimental Mechanics Inc*, volume 77, 2008.

4. D.A. Dederman, D. Burnett, **J.T. Foster**, and J.A. Dykes. In Situ Penetration Testing of Darts with 16-Inch Mobile Gas Gun. In *Proceedings of 24th International Symposium on Ballistics*, number TB149, 2008.

3. J.G. Averett, J.D. Cargile, **J.T. Foster**, and V.K. Luk. Oblique Perforation of Unreinforced Concrete Targets: Experiments and Numerical Simulations. In *Limited Proceedings of 77th Shock and Vibration Conference*, 2007.

2. J.G. Averett, J.D. Cargile, **J.T. Foster**, and V.K. Luk. Projectile Deceleration Due to Perforation Through Layers of Unreinforced Concrete Targets. In *Limited Proceedings of 76th Shock and Vibration Conference*, number U-045. SAVIAC, 2006.

1. **J.T. Foster**, A.A. Barhorst, C.N. Wong, and M.T. Bement. Modeling and Experimental Verification of Frictional Contact-Impact in Loose Bolted Joint Elastic Structures. In *Proceedings of IDETC'05*, number DETC2005-85465. IDETC, 2005.

Technical Reports

10. Peridynamics Capabilities Review Panel. Peridynamics capabilities review panel report. SAND Report 2015-1921, Sandia National Laboratories, Albuquerque, NM and Livermore, CA, 2015.

9. J.V. Cox, G.W. Wellman, J.M. Emery, J.T. Ostien, **J.T. Foster**, T.E. Cordova, T.B. Crenshaw, A. Mota, J.E. Bishop, S.A. Silling, D.J. Littlewood, J.W. Foulk III, K.J. Dowding, K. Dion, B.L. Boyce, J.H. Robbins, and B.W. Spencer. Ductile Failure X-prize. Technical Report SAND2011-6801, Sandia National Laboratories, 2012. doi:10.2172/1029764.

8. E.E. Nishida, **J.T. Foster**, E.W. Klamerus, and D. Burnett. Dynamic behavior of shock isolation/ mitigation materials by kolsky bar experiments. Technical Report SAND2011-8266, Sandia National Laboratories, 2011.

7. **J.T. Foster**, A.E. Fortier, J.G. Averett, J.D. Cargile, V.K. Luk, and D.A. Dederman. Predictive Simulation for Perforation Through Layers of Unreinforced Concrete Targets. Technical Report SAND2008-113, Sandia National Laboratories, 2008.

6. **J.T. Foster**, A.J. Webb, A.E. Fortier, V.K. Luk, and D.A. Dederman. Penetration Code Study for Angle-of-Obliquity (AoO) Experiments into High-Strength Concrete Targets. Technical Report SAND2008-1162, Sandia National Laboratories, 2008.

5. P.D. Coleman, R.A. Bates, M.T. Buttram, S.B. Dron, **J.T. Foster**, R.J. Franco, C.O. Landron, G.M. Loubriel, J.E. Lucero, A. Mar, T.L. Martinez, F.E. Reyes, and B.J. Welch. Void Sensor for Penetrators. Technical Report SAND2007-6528, Sandia National Laboratories, 2007.

4. R.J. Fogler, J.W. Giron, J.A. Jacob, W.P. Wolfe, R.W. Greene, R.D. Tucker, A.E. Fortier, **J.T. Foster**, D.M. Van Zuiden, W.T. O'Rourke, H.D. Nguyen, E. Ollila, and J.R. Phelan. Guided miniature air-deliverable sensor dart. Technical Report SAND2007-6528, Sandia National Laboratories, 2007.

3. **J.T. Foster** and A.J. Webb. Penetration Simulations for Angle-of-Attack (AoA) Experiments into Low Strength Concrete Targets. SAND2007-5256, Sandia National Laboratories, 2007.

2. J.A. Dykes and **J.T. Foster**. Discrete-ULL 1-C Final Test Report. Technical Report SAND2007-4273, Sandia National Laboratories, 2007.

1. **J.T. Foster**. Scale Modeling of Earth Penetrators for In Situ Targets. Technical Report SAND2006-4273, Sandia National Laboratories, 2006.

Technical Presentations

Invited Talks

38. "A finite deformation formulation for reacting porous media with volume change." Oak Ridge National Laboratory, Computational Mechanics Seminar. July 2026.

37. "Applications of Scientific Machine Learning in Petroleum Engineering." Texas A&M University, Harold Vance Department of Petroleum Engineering, Graduate Seminar. November 2025.

36. "Concepts and applications of peridynamics." Babuška Forum Seminar. Oden Institute for Computational Engineering and Science. The University of Texas at Austin. October 2025.

35. "The fundamentals of Generative AI and Machine Learning That Pertain to Geologists and Engineers." AAPG Generative AI, Machine Learning and Analytics for Subsurface Energy. Houston, TX. December, 2024.

34. "Peridynamic modeling of ductile and quasi-brittle fracture." **Plenary**. EUROMECH Colloquium 643: Advances in Peridynamic Modeling. Venice, Italy. September, 2024.

33. "How Researchers Can Collaborate with AI – Panel Discussion" HPC in Energy Workshop. Houston, TX. March, 2024.

32. "Peridynamic modeling of ductile and quasi-brittle fracture." **Conference Plenary**. Africomp 5. Cape Town, SA. November 2022.

31. "What it takes to be an Energy Data Scientist — A Panel Discussion." (with M. Pyrcz). Center for Subsurface Energy & the Environment, The University of Texas at Austin. August 2022.

30. "Applications of SciML in Petroleum Engineering." (Keynote). Energy In Data Conference. Austin, TX February 2022.

29. "Applications of SciML in Petroleum Engineering." XOM Optimization and Data Science Community Meeting. Virtual. February 2022.

28. "Peridynamic modeling of large deformation and ductile fracture" USACM Thematic Workshop on Experimental and Computational Fracture Mechanics. Baton Rouge, LA. February 2020.

27. "Peridynamic modeling of large deformation and ductile fracture: a bond-associated, semi-Lagrangian, constitutive correspondence framework." Tenth International Workshop Meshfree Methods for Partial Differential Equations. Bonn, Germany. September 2019.

26. "Coupling FEM and meshfree peridynamics for the simulation of hydraulic fracturing." 15th National Congress on Computational Mechanics. Austin, TX. August 2019.

25. "Coupling FEM and meshfree peridynamics for the simulation of hydraulic fracturing." Workshop on Meshfree Methods and Advances in Computational Mechanics, In celebration of Professor J.S. Chen's 60th brithday. Dublin, CA. March 2019

24. "Coupling FEM and meshfree peridynamics for the simulation of hydraulic fracturing." USACM Thematic Workshop on Nonlocal Methods in Fracture. Austin, TX, January 2018

23. "Coupling FEM and meshfree peridynamics for the simulation of hydraulic fracturing." Ninth International Workshop Meshfree Methods for Partial Differential Equations. Bonn, Germany. September 2017.

22. "Concepts and applications of peridynamics." Babuška Forum Seminar. Institute for Computational Engineering and Science. The University of Texas at Austin. September 2017.

21. "Finite deformation constitutive models and mechanics of peridynamic mixtures." Workshop on Non-local Material Models and Concurrent Multiscale Methods. Hausdorff Research Institute for Mathematics. Bonn, Germany. April 2017.

20. "Nonlocal models for anamalous transport" Schlumberger EUREKA Fluid Mecahnics Mini-Workshop. Schlumberger-Doll Research Center. July 2016.

19. "Isogeometric peridynamics." USACM Thematic Workshop on Nonlocal Models in Mathematics, Computation, Science, and Engineering. Oak Ridge National Laboratory. October 2015.

18. "A multiphysics model for hydraulic fracture simulation." Eighth International Workshop Meshfree Methods for Partial Differential Equations. Universität Bonn. September 2015.

17. "Nonlocal multiphysics for heterogeneous materials, anomalous diffusion, and fracture." Total. March 2015.

16. "Nonlocal multiphysics for heterogeneous materials, anomalous diffusion, and fracture." Graduate Aerospace Laboratories, California Institute of Technology. January 2015.

15. "Nonlocal multiphysics for heterogeneous materials, anomalous diffusion, and fracture." Institute for Computational Engineering Science, The University of Texas at Austin. October 2014.

14. "Nonlocal multiphysics for heterogeneous materials, anomalous diffusion, and fracture." University of Illinos – Urbana-Champaign, Department of Aerospace Engineering. September 2014.

13. "Nonlocal multiphysics for heterogeneous materials, anomalous diffusion, and fracture." Center for Mechanics of Solids, Structures and Materials, The University of Texas at Austin, Department of Aerospace Engineering and Engineering Mechanics. September 2014.

12. "Nonlocal multiphysics for heterogeneous materials, anomalous diffusion, and fracture." ExxonMobil - Corporate Strategic Research. July 2014.

11. "A model for nonlocal diffusion and fluid-driven fracture." USACM/IUTAM Symposium on Connecting Multiscale Mechanics to Complex Material Design. Northwestern University. May 2014.

10. "Nonlocal multiphysics for heterogeneous materials, anomalous diffusion, and fracture." The University of Texas at Austin, Department of Petroleum & Geosystems Engineering. March 2014.

9. "Nonlocal multiphysics for heterogeneous materials, anomalous diffusion, and fracture." Northwestern University, Department of Mechanical Engineering. January 2014.

8. "Peridynamics as a unified theory for heterogenous media, anomalous porous flow, and fracture." The University of Texas at Austin, Department of Petroleum & Geosystems Engineering. October 2013.

7. "Unifying the mechanics of continuous and discontinuous media." Army Research Laboratory. February 2013.

6. "Unifying the mechanics of continuous and discontinuous media." The Johns Hopkins University, Center for Advanced Ceramics and Metallic Systems. July 2012.

5. "Unifying the mechanics of continuous and discontinuous media." Texas Tech University, Mechanical Engineering. April 2012.

4. "Hydraulic fracturing and its environmental impact: a short address of major public concerns." Presentation for the Center for Simulation, Visualization, and Real-Time Prediction participation in UTSA Earthweek 2012. April 2012.

3. "Unifying the mechanics of continuous and discontinuous media." 2011 International Workshop on Intensive Loading and its Effects. State Key Laboratory of Explosion Science and Technology, Beijing Institute of Technology. Beijing, China. December 2011.

2. "Peridynamic modeling of viscoplasticity and dynamic fracture." University of Nebraska, Engineering Mechanics. April 2010.

1. "Peridynamic modeling of viscoplasticity and dynamic fracture." University of New Mexico, Mechanical Engineering. February 2010.

Conferences

42. "Revisiting finite deformation poromechanics: deriving a nonlinear Biot coefficient from first principles." 20th U.S. National Congress on Theoretical and Applied Mechanics (USNCTAM 2026). Pasadena, CA. June 2026.

41. "The peridynamic Jacobian." 18th National Congress on Computational Mechanics. USACM. Chicago, IL. July 2025.

40. "The Rancher, the Hunter, the Renaissance Man" 16th World Conference on Computational Mechanics. Vancouver, BC. July, 2024.

39. "Lessons learned in 15 years of peridynamics correspondence modeling" Quarter Century of Peridynamics. Tucson, AZ. April, 2024.

38. "Multi-Output Physics-Informed Neural Networks for Forward and Inverse PDE Problems with Uncertainties" (with M. Yang) 17th U.S. National Congress on Computational Mechanics. Albuquerque, NM. July, 2023.

37. "A machine-learning framework for peridynamic material models with physical constraints." (with X. Xu and M. D'Elia) U.S. National Congress on Theoretical and Applied Mechanics. Austin, TX. June 2022.

36. "Applications of SciML in Petroleum Engineering" AAPG Energy In Data Conference. Austin, TX. February, 2022.

35. "Coupling FEM and meshfree peridynamics for the simulation of hydraulic fracturing." (with J.R. York) World Congress on Computational Mechanics XIII. July 2018.

34. "Coupling FEM and meshfree peridynamics for the simulation of hydraulic fracturing." (with J.R. York) ECCOMAS 6th European Conference on Computational Mechanics. Glasgow, UK. June 2018.

33. "A Hypoelastic Constitutive Correspondence Model for Peridynamics." (with M. Behzadnasab, X. Xu) US National Congress on Theoretical and Applied Mechanics. June 2018.

32. "A Finite Deformation Generalized Correspondence Theory for Peridynamic Material Modeling" (with X. Xu). ASME 2017 International Mechanical Engineering Congress and Exposition. November 2017.

31. "Coupling FEM and meshfree peridynamics for the simulation of hydraulic fracturing." (with J.R. York) Ninth International Workshop Meshfree Methods for Partial Differential Equations. September 2017.

30. "Finite Deformation Constitutive Models and Mechanics of Peridynamic Mixtures." (with X. Xu). 14th US National Congress on Computational Mechanics. July 2017.

29. "Modeling hydraulic fracturing with a pressure dependent cap model and peridynamics." (with J.R. York). EMI 2017. June 2017.

28. "A model for the transport of miscible fluids in the presence of anomalous diffusion." 2017 SIAM Conference on Computational Science and Engineering. February 2017.

27. "A finite deformation generalized correspondence theory for peridynamic material modeling" (with X. Xu). ASME 2016 International Mechanical Engineering Congress and Exposition. November 2016.

26. "A model for the transport of miscible fluids in the presence of anomalous diffusion." (with R. Tabasi). USACM Thematic Workshop on Isogeometric Analysis and Meshfree Methods. October 2016.

25. "A variationally consistent approach to constrained motion." 24th International Congress on Theoretical and Applied Mechanics. August 2016.

24. "A model for the transport of miscible fluids in the presence of anomalous diffusion." (Keynote, with R. Tabasi). World Congress on Computational Mechanics XII. July 2016.

23. "A peridynamic model for hydraulic fracture." (with H. Ouchi, J.R. York, M.D. Brothers, M.M. Sharma). SIAM Annual Conference. July 2016.

22. "A peridynamic model for hydraulic fracture." (with H. Ouchi, J.R. York, M.M. Sharma). Engineering Mechanics Institute Conference 2016. May 2016.

21. "Bending Failure in Peridynamic Plates." (with J. O'Grady). ASME 2015 International Mechanical Engineering Congress and Exposition. November 2015.

20. "Mesoscale Simulations Investigating the Effects of Shock Wave Stability in Granular Materials with Peridynamics." (with R. Rahman, A. Peterson, T. Vogler). 13th US National Congress on Computational Mechanics. July 2015.

19. "Regularizing numerical simulations of shear-banding using a peridynamics-based plasticity formulation." (with Md.I.H. Kahn). ASME 2014 International Mechanical Engineering Congress and Exposition. November 2014.

18. "An Ordinary State Based Plasticity Model For Peridynamics." (with J.A. Mitchell). ASME 2014 International Mechanical Engineering Congress and Exposition. November 2014.

17. "Fracture in plates and shells with peridynamic non-ordinary state-based models." Meshfree Methods for Large-Scale Computational Science and Engineering. October 2014.

16. "An Overview of the Progress of Meshfree Particle Methods: From SPH to EFG to RKPM to Meshfree Peridynamics." (with W.K. Liu, M. Bessa). Meshfree Methods for Large-Scale Computational Science and Engineering. October 2014.

15. "A nonlocal poroelastic approach to fluid driven fracture." (with J.R. York, A. Katiyar, H. Ouchi, M. Sharma). World Congress on Computational Mechanics XI. July 2014.

14. "Reproducing Continuum Dynamics". (with M. Bessa, W.K. Liu, T. Belytschko). World Congress on Computational Mechanics 2014. July 2014.

13. "A nonlocal poroelastic approach to fluid driven fracture." (with J.R. York, A. Katiyar, H. Ouchi, M. Sharma). US National Congress on Theoretical and Applied Mechanics. June 2014.

12. "Bridging the length scales by linking the atomistic model with coarser peridynamic models through molecular dynamics simulation of Polyethylene". (with R. Rahman). Mach Conference 2014. April 2014.

11. "Regularizing numerical simulations of strain-localization using a peridynamics-based plasticity formulation". (with Md.I. Kahn, D.J. Littlewood, and J.A. Mitchell). International Workshop on Computational Mechanics of Materials, IWCMM XXIII. October 2013

10. "A non-local formulation for fluid flow and mass transport in porous media based on peridynamic theory". (with A. Katiyar and M. Sharma). 12th US National Congress on Computational Mechanics. July 2013.

9. "A novel hierarchical multiscale modeling framework for polyethylene systems using Peridynamics and molecular dynamics". (with R. Rahman). 2013 Mach Conference, Annapolis, MD. April 2013.

8. "Two-Dimensional Semi-Analytic Solutions to the Linearized State-Based Peridynamic Equilibrium Equation". (with J.T. O'Grady). USACM Workshop on Nonlocal Damage and Failure: Peridynamics and other nonlocal methods. March 2013.

7. "A Peridynamics Based Hierarchical Multiscale Modeling Framework Between Continuum and Atomistic Scales". (with R. Rahman, A. Haque). USACM Workshop on Nonlocal Damage and Failure: Peridynamics and other nonlocal methods. March 2013.

6. "Lessons Learned in Modeling Ductile Failure with Peridynamics". (with D.J. Littlewood). USACM Workshop on Nonlocal Damage and Failure: Peridynamics and other nonlocal methods. March 2013.

5. "A Peridynamics Formulation of the Coupled Mechanics-Fluid Flow Problem". (with A. Katiyar, H. Ouchi, M.M. Sharma). USACM Workshop on Nonlocal Damage and Failure: Peridynamics and other nonlocal methods. March 2013.

4. "Implicit time integration of an ordinary state-based peridynamic plasticity model with isotropic hardening." (with D.J. Littlewood, J.A. Mitchell, M.L. Parks). ASME IMECE 2012. November 2012.

3. "Implicit time integration of an ordinary state-based peridynamic plasticity model with isotropic hardening." (with D.J. Littlewood, J.A. Mitchell, M.L. Parks). SiViRT Simulation and Vizualization Symposium. November 2012.

2. "Peridynamic Modeling of Localization in Ductile Metals." (with D.J. Littewood and B.L. Boyce) International Workshop on Computational Mechanics of Materials IWCMM XXII. September 2012.

1. "Viscoplasticity using peridynamics." (with S.A. Silling and W. Chen) 10th US National Congress on Computational Mechanics. July 2009.

Student Delivered

18. "Topology optimization for fracture resistance using differential programming and peridynamics." (with M. Schmitz) 20th U.S. National Congress on Theoretical and Applied Mechanics (USNCTAM 2026). Pasadena, CA. June 2026.

17. "Dynamic fracture modeling of ductile materials with peridynamics." (with M. Behzadnasab) 13th World Congress on Computational Mechanics / 2nd Pan American Congress on Computational Mechanics (WCCM 2018). New York City, New York. July 2018.

16. "Peridynamic modeling of dynamic fracture in metallic materials." (with M. Behzadnasab) USACM Thematic Conference on Meshfree and Particle Methods: Applications and Theory (MFPM 2018). Santa Fe, New Mexico. September 2018.

15. "A peridynamic study of predicting ductile fracture in additively manufactured metal." (with M. Behzadnasab) The 1st Annual Meeting of SIAM Texas-Louisiana Section. Baton Rouge, Louisiana. October 2018.

14. "On the stability of the generalized, ordinary, finite deformation constitutive correspondence model of peridynamics." (with M. Behzadnasab) ASME's International Mechanical Engineering Congress & Exposition (IMECE 2018). Pittsburgh, Pennsylvania, November 2018.

13. "A stabilized hypoelastic constitutive correspondence model for peridynamics." (with M. Behzadnasab) 2019 Minerals, Metals & Materials Society Annual Meeting & Exhibition (TMS 2019). San Antonio, Texas. March 2019.

12. "Dynamic ductile fracture characterization with peridynamics: A Sandia Fracture Challenge study." (with M. Behzadnasab) The Fourteenth Pan American Conference on Applied Mechanics (PACAM XVI). Ann Arbor, Michigan. May 2019.

11. "A model for the transport of miscible fluids in the presence of anamalous diffusion." (with R. Tabasi). ASME 2017 International Mechanical Engineering Congress and Exposition. November 2017.

10. "Modeling perturbed shock wave decay in granular materials with intra-granular fracture." (with M. Behzadinasab and T.J. Vogler) 20th Biennial Intl. Conference of the APS Topical Group on Shock Compression of Condensed Matter. July 2017.

9. "Modeling of Contact and Non-Local Friction in a Peridynamic Framework." (with J.R. York). 13th US National Congress on Computational Mechanics. July 2015.

8. "Mesh-Free Non-ordinary Peridynamic Bending." (with J. O'Grady). 13th US National Congress on Computational Mechanics. July 2015.

7. "A Peridynamic Model for Hydraulic Fracture." (with H. Ouichi, A. Katiyar, M. Sharma). 13th US National Congress on Computational Mechanics. July 2015.

6. "Peridynamic beams, plates, and shells: a non-ordinary state-based model." (with J. O'Grady). ASME 2014 International Mechanical Engineering Congress and Exposition. November 2014.

5. "Peridynamic beams, plates, and shells: a non-ordinary state-based model." (with J. O'Grady). Society of Engineering Science 2014. October 2014.

4. "The Next Generation Model for Predicting the Growth of Complex Fracture Networks." (with J.R. York). 2014 Hydraulic Fracturing and Sand Control Joint Industry Program Technical Review. April 2014.

3. "A peridynamic model of diffusive fluid flow through a deformable media." (with J.R. York). 2013 SACNAS National Conference. October 2013.

2. "A complex-step method for tangent-stiffness calculation in a massively parallel computational peridynamics code." (with M.D. Brothers and H.R. Millwater). 12th US National Congress on Computational Mechanics. July 2013.

1. "Intragranular fracture and frictional effects in granular materials under pressure-shear loading." (with A.M. Peterson and T.J. Vogler) 18th Biennial Intl. Conference of the APS Topical Group on Shock Compression of Condensed Matter held in conjunction with the 24th Biennial Intl. Conference of the Intl. Association for the Advancement of High Pressure Science and Technology (AIRAPT). July 2013.

Poster

6. "Sandia Fracture Challenge 2017: Peridynamics blind prediction of dynamic crack growth.' (with M. Behzadnasab) 13th World Congress on Computational Mechanics / 2nd Pan American Congress on Computational Mechanics (WCCM 2018). New York City, New York. July 2018.

5. "Sandia Fracture Challenge 2017: peridynamic blind prediction of dynamic crack growth in ductile materials." (with M. Behzadnasab) USACM Thematic Conference on Meshfree and Particle Methods: Applications and Theory (MFPM 2018). Santa Fe, New Mexico. September 2018.

4. "A stabilized, hypoelastic constitutive correspondence framework for peridynamics." (with M. Behzadnasab) SIAM Conference on Computational Science and Engineering 2019 (SIAM-CSE19). Spokane, Washington. February 2019.

3. "Modeling perturbed shock wave decay in granular materials with intra-granular fracture." (with M. Behzadinasab and T.J. Vogler) 20th Biennial Intl. Conference of the APS Topical Group on Shock Compression of Condensed Matter. July 2017.

2. "A Peridynamic Model for Hydraulic Fracture." (with J.R. York) USACM Thematic Workshop on Nonlocal Models in Mathematics, Computation, Science, and Engineering. Oak Ridge National Laboratory. October 2015.

1. "Intragranular fracture and frictional effects in granular materials under pressure-shear loading." (with A.M. Peterson and T.J. Vogler) 18th Biennial Intl. Conference of the APS Topical Group on Shock Compression of Condensed Matter held in conjunction with the 24th Biennial Intl. Conference of the Intl. Association for the Advancement of High Pressure Science and Technology (AIRAPT). July 2013.

Software

Core developer for *Peridigm* open source peridynamics software

Website: https://peridigm.sandia.gov

Source Code: https://github.com/peridigm/peridigm

Various other projects developed and contributed to

Source Codes: https://github.com/johntfoster

Blog

Contains various useful code snippets, examples, and resources primarily targeted to assist in students working under my supervision and other researchers in scientific computation.

Website: https://johnfoster.pge.utexas.edu/blog/

Grant Proposals

Funded

Role and Co-Investigators	Title	Agency	Grant Total (My Share)	Grant Period
PI	Sandia X-Prize Necking Challenge	Sandia National Laboratories	$44,700 ($44,700)	1/2012-12/2012
PI	Peridynamic Simulation of Granular Materials Undergoing Shock Compression	Sandia National Laboratories	$32,597 ($32,597)	1/2012-12/2012
PI	Statistical coarse-graining of molecular dynamics into peridynamics	Subaward from ARL via Johns Hopkins University	$91,925 ($91,925)	1/2012-12/2012
co-PI, Sharma, M. (PI)	Fracture Design, Placement And Sequencing In Horizontal Wells, DE-FOA-0000724	National Energy Technology Laboratory	$1,592,451 ($275,250)	1/2012-12/2016
PI	Application of Peridynamics to Hydraulic Fracture Modeling	UTSA VPR	$18,927 ($18,927)	9/2012-8/2013
PI	Peridynamic simulation of pressure-shear experiments on granular media	Sandia National Laboratories	$29,071 ($29,071)	1/2013-12/2013
PI	Towards a multiscale failure modeling paradigm for polymers: statistical coarse-graining of molecular dynamics into peridynamics.	Subaward from ARL via The Johns Hopkins University	$91,925 ($91,925)	1/2013-12/2013
PI	Fiber failure modeling with peridynamics	Subaward from ARL via The Johns Hopkins University	$101,306 ($101,306)	1/2013-12/2013
PI	Predictive simulation of material failure using peridynamics-advanced constitutive modeling, verification, and validation, BAA-AFOSR-2012-0001	AFOSR	$360,000 ($360,000)	9/2013-12/2015
co-PI, Madenci, E. (PI),Bobaru, F. (co-PI), Chawla, N. (co-PI), Du, Q. (co-PI)	MURI Center for Material Failure Prediction Through Peridynamics, ONRBAA12-020	AFOSR	$7,500,000 ($959,153)	9/2013-12/2018
co-PI, Sharma, M. (PI)	GRA support for hydraulic fracture modeling with peridynmaics (Jason York)	IAP on Hydraulic Fracturing	$39,819 ($39,819)	1/2015-12/2015

PI	Nonlocal and fractional order methods for near-wall turbulence, large-eddy simulation, and fluid-structure interaction, ONRFOA14-012	ARO	$345,000 ($345,000)	1/2015-12/2018
co-PI, Sharma, M. (PI)	GRA support for hydraulic fracture modeling with peridynmaics (Jason York)	IAP on Hydraulic Fracturing	$33,845 ($33,845)	1/2015-12/2015
PI	Pulse Fracture Simulation	GE Global Research	$100,000 ($100,000)	1/2016-12/2016
co-PI, Sharma, M. (PI)	GRA support for hydraulic fracture modeling with peridynmaics (Shivam Agrawal)	IAP on Hydraulic Fracturing	$67,664 ($33,832)	1/2016-12/2019
PI	ASCeND: ASymptotically Compatible strong form foundations for Nonlocal Discretization.	Sandia National Laboratories	$300,000 ($300,000)	9/2018-10/2021
PI	Student and Postdoc Travel Support for the 15th USNCCM in Austin, TX	NSF, Proj. No. 1935320.	$25,000 ($25,000)	1/2019-8/2019
co-PI, Pyrcz, M. (co-PI)	Hildebrand Seed Grant for Data Science Research Initiative	PGE	$50,000 ($25,000)	1/2019-12/2019
PI	Moncrief Grand Challenge: GFEM Framework for Reservoir Simulation of Unconventionals	Oden Institute	$75,000 ($75,000)	1/2019-9/2019
co-PI, Pyrcz, M. (co-PI)	DiReCT: Digital Reservoir Characterization Technology	DiReCT IAP	$1,200,000 ($400,000)	7/2019-9/2024
PI	MATNIP: Mathematical Foundations of Nonlocal Interface Problems	Sandia National Laboratories	$121,000 ($121,000)	9/2020-10/2022
PI	Assessing Capillary End Effects on Large Scale Tight Reservoir Drainage	American Chemical Society	$110,000 ($110,000)	1/2021-12/2023
co-PI, Espinoza, D (PI)	Hydrogen storage in salt caverns in the Permian Basin: Seal integrity evaluation and field test	DOE	$1,854,361 ($122,000)	9/2023-10/2025
co-PI, Lu, Y. (PI), Pyrcz, M. (co-PI)	Unconventional Well Optimization based on Machine Learning	University Lands	$75,000 ($25,000)	9/2023-8/2025
co-PI, Lu, Y. (PI), Pyrcz, M. (co-PI)	Unconventional Well Optimization based on Machine Learning	Hildebrand Seed Grant	$75,000 ($25,000)	9/2023-8/2025
co-PI, Espinoza, D (co-PI), Hu, M. (PI) et al.	Cyclic Injection for Commerical Seismic-Safe Geologic H2 Production	ARPA-E	$521,835 ($279,819)	9/2024-9/2026

PI	SciML at CAMINO	Sandia National Laboratories	$525,000 ($525,000)	10/2024-9/2027
co-PI, Espinoza, D (PI)	Fundamental Understanding of Thermo-hydraulic Fractures in Porous Media for Improved Geothermal Energy	DOE	$100,000 ($50,000)	9/2023-10/2025
		Totals	$15,481,426 ($4,640,169)	

Courses Taught

PGE 310 - Numerical Methods and Programming (UT S2020-24)

PGE 334 - Reservoir Geomechanics (UT S2015, S2016, S2017, S2018)

PGE 383 - Advanced Geomechanics (UT F2014, F2015, S2025)

PGE 323M - Reservoir Engineering III (UT F2015-24)

PGE 392K - Reservoir Simulation (UT F2025)

Introduction to High-Performance Computing (UTSA F2012, F2013, S2014, UT S2018, F2018, S2019-23)

ME 6043 – Continuum Mechanics (UTSA F2012, F2014)

ME 4603 – Finite Element Analysis (UTSA F2011)

ME 400/500 – Numerical Methods (UNM F2010)

Advising and related student services

Graduate Students (Graduated)

PhD

1. Elnara Rustamzade PhD PGE 2026 *co-advised with M. Pyrcz*
2. Mingyaun Yang PhD PGE 2022
3. Rambod Yousefzadeh Tabasi PhD EM 2022
4. Xiao Xu PhD CSEM 2022
5. Mayowa Olugbenga PhD PE 2021 *co-advised with K. Gray*
6. Yu Leng PhD PE 2020 (Postdoctoral Researcher at LANL)
7. Shivam Agrawal, PhD PE 2019 *co-advised with M. Sharma* (SparkCognition)

8. Masoud Behzadinasab PhD EM 2019 (PTC)
9. Jason York, PhD PE 2018 (Artemis Capital)
10. James O'Grady, PhD ME 2014 (Army Research Lab)

MS

1. Amena Alelg, MS PGE 2026
2. Muhannad Alabdullateef, MS PGE 2024
3. Fehmi Ozbayrak, MS PGE 2024 *co-advised with M. Pyrcz*
4. Nelson Morrow, MS EM 2024
5. Ruoyu Wang, MS PE 2023 *co-advised with L. Lake & M. Pyrcz*
6. Odai Elyas, MS PE 2022
7. Akhil Potla, MS EM 2022 *co-advised with L. Lake*
8. Katy Hanson, MS PE 2018 *co-advised with E. van Oort*
9. Xiao Xu, MS PE 2017
10. Eric Lynd, MS PE 2017 *co-advised with Q. Nguyen*
11. Sai Uppati, MS PE 2016
12. Amanda Peterson, MS ME 2014 (UTSA)
13. Md. Imran Khan, MS ME 2014 (UTSA)
14. Michael Brothers, MS ME 2013 (UTSA)
15. Jason York, MS ME 2012 (UTSA)
16. Arron Werthiem, MS ME 2012 (UTSA)

Graduate Students (In Progress)

PhD

1. Shaika Aldossary (PGE)
2. Syed Talha Tirmizi (PGE) *co-advised with M. Hesse*
3. Muhannad Alabdullateef (PGE)
4. Dinghan Wang (PGE) *co-advised with M. Pyrcz & Y. Lu*

MS

1. Saris Visessumon (PGE) *co-advised with M. Pyrcz*
2. Dursun Dashdamirov (PGE) *co-advised with M. Pyrcz*
3. Emmanuel Ekpesu (PGE)

Postdoctoral Researcher's Supervised

1. Madison Schmitz, Ph.D. (UT)
2. James O'Grady, Ph.D. (UT)
3. Rezwanur Rahman, Ph.D. (UTSA/UT)
4. Shamima Yasmin, Ph.D. (UTSA)

Undergraduate Research Assistants

1. P. Eric Briseno, B.S.M.E. 2013
2. Robert Knobles, B.S.M.E. 2014 (Baker-Hughes)
3. Robert Brothers
4. Jason Crandall
5. Sam Petzold – Moncrief Summer Intern

Academic-related Professional and Public Service

Conferences/Workshops Organized

1. US National Congress on Computational Mechanics 15
   Conference Chair
   Held in Austin, TX, July 28-August 1, 2019
   http://15.usnccm.org
2. Workshop on Nonlocal Methods in Fracture
   Sponsered by the US Association for Computational Mechanics.
   Held in Austin, TX, Janruary 15-16, 2018
   http://nmf2018.usacm.org
3. Workshop on Isogeometric Analysis and Meshfree Methods
   Sponsered by the US Association for Computational Mechanics.

Held at UCSD, October 10-12, 2016

http://iga-mf.usacm.org

4. Workshop on Meshfree Methods for Computational Science and Engineering

   Sponsered by the US Association for Computational Mechanics.

   Held at UCF, October 27-28, 2014

   http://mmlcse2014.usacm.org

5. Workshop on Nonlocal Damage and Failure: Peridynamics and other nonlocal models.

   Sponsered by the US Association for Computational Mechanics.

   Held at UTSA Downtown Campus, March 11-12, 2013

   http://ndf2013.usacm.org

Administrative and Committee Service

Committee Assignments

Department

PGE Undergraduate Studies 2015-2025

PGE Graduate Admissions Committee 2015-2017

PGE Department Awards Committee 2014-2017

Graduate Committee 2013-2014 (UTSA)

Faculty Search Committee 2013-2014 (UTSA)

Department Promotional Activities 2012-2013 (UTSA)

Seminar 2011-2012 (UTSA)

University

Cockrell School Math Committee 2025

Cockrell School Degrees and Courses 2020-2025

Cockrell School Engineering Honors 2015-2019

Undergraduate Research Day Planning Committee 2013-2014 (UTSA)

Student Organization Advisor

Programming for Engineers & Scientists 2016

Tau Beta Pi 2013-2014 (UTSA)

Formula SAE Car Team 2013-2014 (UTSA)

Associate Editor

Journal of Peridynamics and Nonlocal Modeling

Reviewer For

Journals

Computational Geosciences, Journal of Applied Mechanics, Computational Methods in Applied Mechanics and Engineering, Journal of Computational Particle Mechanics, Journal of Microelectromechanical Systems, Computational Mechanics, Int. Journal of Fracture, Applied Mathematics & Computation, Int. Journal of Impact Engineering, Engineering Fracture Mechanics, Experimental Mechanics, Review of Scientific Instruments, Int. Journal of Multiscale Computational Engineering, Int. Journal of Solids and Structures, CMC: Computers, Materials, & Continua, Journal of Mechanics of Materials and Structures.

Books

Split Hopkinson (Kolsky) Bar. W. Chen and B. Song. Springer 2010.

Book Proposals

CRC Press

Organizations

Society of Petroleum Engineers, US Association for Computational Mechanics, Pi Tau Sigma - Mechanical Engineering Honor Fraternity, Tau Beta Pi - National Engineering Honor Society, American Society of Mechanical Engineers, American Institute of Aeronautics and Astronautics, Society for Experimental Mechanics – Dynamic Behavior of Materials Technical Division Committee Member, DYMAT, American Society for Engineering Education

Vita

John T. Foster is a professor in the Hildebrand Department of Petroleum and Geosystems Engineering and Aerospace Engineering and Engineering Mechanics (by courtesy) at the University of Texas at Austin. He is co-founder and CTO of Daytum. He received his BS and MS in mechanical engineering from Texas Tech University and PhD from Purdue University. He is a registered Professional Engineer in the State of Texas. During his career in research he has been involved in many projects ranging from full scale projectile penetration field tests, to laboratory experiments using Kolsky bars, to modeling and simulation efforts using some of the world's largest computers. His research interests are in experimental and computational mechanics and multi-scale modeling with applications to geomechanics, impact mechanics, fracture mechanics, and anomalous transport processes. Additionally, he has interest in fundamental theoretical advancement of the peridynamic theory of solid mechanics. His teaching interests are in all areas of theoretical and computational mechanics.